Illustration 15.2 C++ Program to demonstrate function overloading

```
#include <iostream>
using namespace std;
long add(long, long);
long add(long,long,long);
float add(float, float);
int main()
{
 long a, b, c,d;
 float e, f, g;
 cout << "Enter three integers\n";
cin >> a >> b>>c;
d=add(a,b,c);          //number of arguments different  but  same data type
cout << "Sum of 3 integers: " << d << endl;
cout << "Enter two integers\n";
cin >> a >> b;
c = add(a, b);         //two arguments with same data type
cout << "Sum of 2 integers: " << c << endl;
cout << "Enter two floating point numbers\n";
 cin >> e >> f;
 g = add(e, f);        //two arguments with different data type
cout << "Sum of floats: " << g << endl;
}
long add(long c, long g)
{
 long sum;
 sum = c + g;
 return sum;
}
float add(float c, float g)
{
 float sum;
 sum = c + g;
 return sum;
}
long add(long c, long g,long h)
{
 long sum;
 sum = c + g+h;
 return sum;
}
```

Output

```
Enter three integers
3 4 5
Sum of 3 integers: 12
Enter two integers
4 6
Sum of 2 integers: 10
Enter two floating point numbers
2.1 3.1
Sum of floats: 5.2
```


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